Exercise 3: Text Summarization Using Different Prompting Techniques

Aim

To design zero-shot, one-shot, and few-shot prompts to summarize an article into 50 words and compare the summaries based on accuracy, completeness, and readability.

Algorithm

- Store a sample article.
- Create zero-shot, one-shot, and few-shot summarization prompts.
- Specify a 50-word summary limit.
- Display the prompts.
- Compare the summaries generated by an LLM for accuracy, completeness, and readability.

Python Code

```
Article: {article}
"""

one_shot = f"""
Example:
Article: Trees absorb carbon dioxide and release oxygen.
They also provide habitats and prevent soil erosion.

Summary: Trees support cleaner air, provide wildlife habitats,
and help prevent soil erosion.

Now summarize this article in 50 words:
{article}
"""

few_shot = f"""
Example 1:
Article: Exercise improves fitness and supports mental health.
Summary: Exercise benefits physical fitness and mental well-being.

Example 2:
Article: Solar power converts sunlight into electricity.
Summary: Solar energy produces electricity from sunlight.

Task: Summarize the following article in 50 words.
Preserve important facts and use clear language.

Article: {article}
"""
```

Sample Output

```
ZERO-SHOT PROMPT

Summarize the following article in exactly 50 words.
Preserve the main ideas and avoid adding new information.

Article:
Artificial Intelligence is transforming healthcare through
medical image analysis, disease prediction, and personalized
treatment recommendations. AI systems can help doctors detect
diseases earlier and analyze large amounts of medical data.
However, privacy, bias, cost, and the need for human supervision
remain important challenges. AI supports healthcare professionals
but should not completely replace their judgment.



ONE-SHOT PROMPT

Example:
Article: Trees absorb carbon dioxide and release oxygen.
They also provide habitats and prevent soil erosion.

Summary: Trees support cleaner air, provide wildlife habitats,
and help prevent soil erosion.

Now summarize this article in 50 words:

Artificial Intelligence is transforming healthcare through
medical image analysis, disease prediction, and personalized
treatment recommendations. AI systems can help doctors detect
diseases earlier and analyze large amounts of medical data.
However, privacy, bias, cost, and the need for human supervision
remain important challenges. AI supports healthcare professionals
```

Result

Three summarization prompts were created. Their generated summaries can be evaluated based on accuracy, completeness, and readability. The article above is a sample article for this experiment.